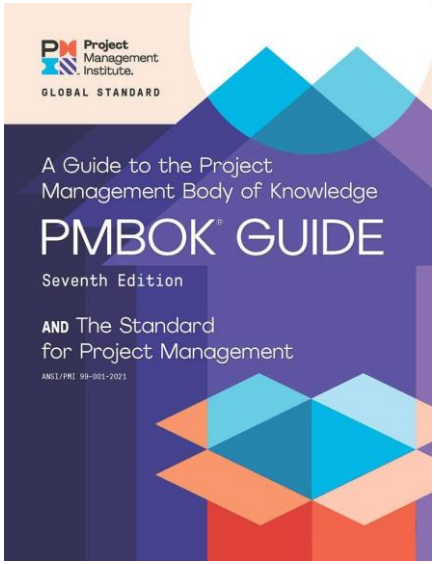
IT3120- Lecture 8

Project Management - PM Lifecycle and PM Process groups

PM Lifecycle and Process groups



Lecture Objectives

- Explain the concept of the project lifecycle
- Identify and describe PM process groups
- Understand how lifecycle and process groups interact
- Apply PMBOK 7 principles to IT projects
- Relate theoretical concepts to real-world IT project scenarios

What is a Project?

“A temporary endeavor undertaken to create a unique product, service, or result.”



A project is a **time-bound effort to create something new that delivers value**.

Key Characteristics of a Project



Temporary Nature

A project has a **defined beginning and end**. Not ongoing work — it ends once objectives are met.



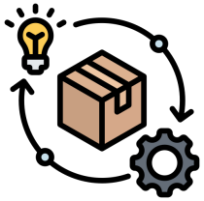
Unique Output

Every project delivers something **unique**, such as:

- A software application
- A new system feature
- An IT infrastructure setup

Even repeated projects differ in conditions, stakeholders, or requirements.

Key Characteristics of a Project



Progressive Elaboration

Project details are **developed step by step** as more information emerges — common in evolving IT requirements.



Value Delivery Focus (PMBOK 7 Emphasis)

PMBOK 7: success means **delivering value** to stakeholders, not just outputs.

Project Management



Applying knowledge, skills, tools, and techniques to meet requirements and deliver value.



Project management is **how you plan, execute, and control a project to achieve its goals**.

Core Concepts of Project Management

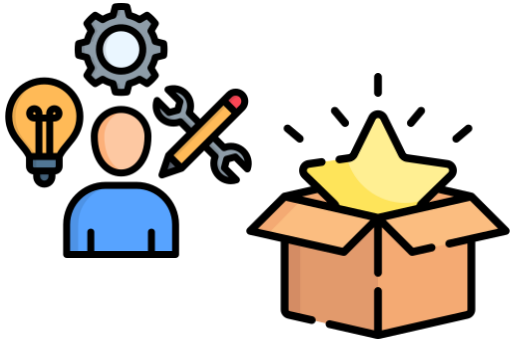
1. Application of Capabilities

Project management involves using appropriate:

- Knowledge (methods, frameworks)
- Skills (leadership, communication)
- Tools & techniques (scheduling tools, risk analysis)

2. Focus on Value Delivery

PMBOK 7: it's about **delivering value** to stakeholders — not just completing tasks.



Core Concepts of Project Management

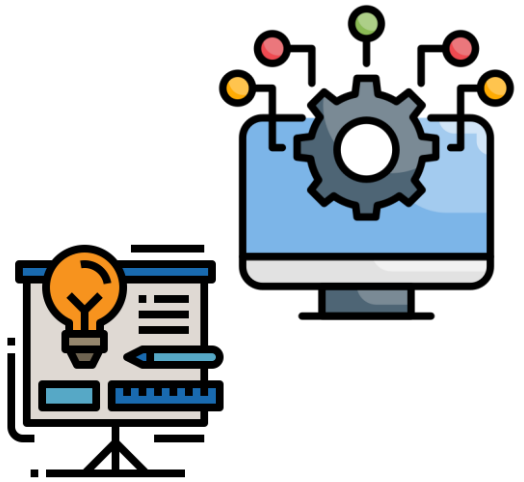
3. System Thinking Approach

Projects are part of a larger system — decisions in one area can affect others.

4. Tailoring

Project management practices should be **adapted** based on:

- Project size
- Complexity
- Industry (e.g., IT, construction)



The 12 Project Management Principles

The 12 Project Management Principles (PMBOK® Guide) are the foundation of modern PM practice.

They shift the focus from rigid processes to a flexible, value-driven, people-centered approach — key for IT projects.

The 12 Project Management Principles

1. **Be a Diligent, Respectful, and Caring Steward**

   Act with **integrity, accountability, and ethical responsibility** — manage resources wisely and respect stakeholders.

2. **Create a Collaborative Project Team Environment**

   A successful project depends on **team collaboration and trust** — foster open communication so the team can thrive.

The 12 Project Management Principles

3. **Effectively Engage with Stakeholders**

   Projects succeed when stakeholders are **actively involved and their expectations are managed**. Identify them early and keep them engaged throughout.

4. **Focus on Value**

   The ultimate goal of any project is to **deliver value**, not just outputs — benefits must justify the investment.

The 12 Project Management Principles

5. **Recognize, Evaluate, and Respond to System Interactions**

   Projects operate within a **larger organizational and environmental system**. Decisions in one area can affect others — take a holistic view.

6. **Demonstrate Leadership Behaviors**

   Effective project managers demonstrate **leadership qualities** — motivation, decision-making, and emotional intelligence to guide the team.

The 12 Project Management Principles

7. **Tailor Based on Context**

   No one-size-fits-all approach — adapt practices to size, complexity, risk, and culture.

8. **Build Quality into Processes and Deliverables**

   Quality should be **proactively built into the project processes**, not just inspected at the end — this reduces rework and defects.

The 12 Project Management Principles

9. **Navigate Complexity**

   Projects involve uncertainty and change — the need to **understand and manage complexity** using the right tools and adaptive thinking.

10. **Optimize Risk Responses**

    Project managers should **identify, assess, and respond to risks continuously** — to minimize threats and seize opportunities.

The 12 Project Management Principles

11. Embrace Adaptability and Resilience

Stay flexible and resilient — adapt plans and recover quickly from disruption.

12. Enable Change to Achieve the Envisioned Future State

Projects are vehicles for change — focuses on **managing and facilitating change effectively** toward the desired future state.

In-Class Activity 1 – Multiple Choice Question

Which concept emphasizes delivering benefits and usefulness to stakeholders?

A. Temporary

B. Progressive development

C. Unique output

D. Focus on value



Project Lifecycle Overview

The Project Lifecycle is the series of phases a project goes through from start to finish.

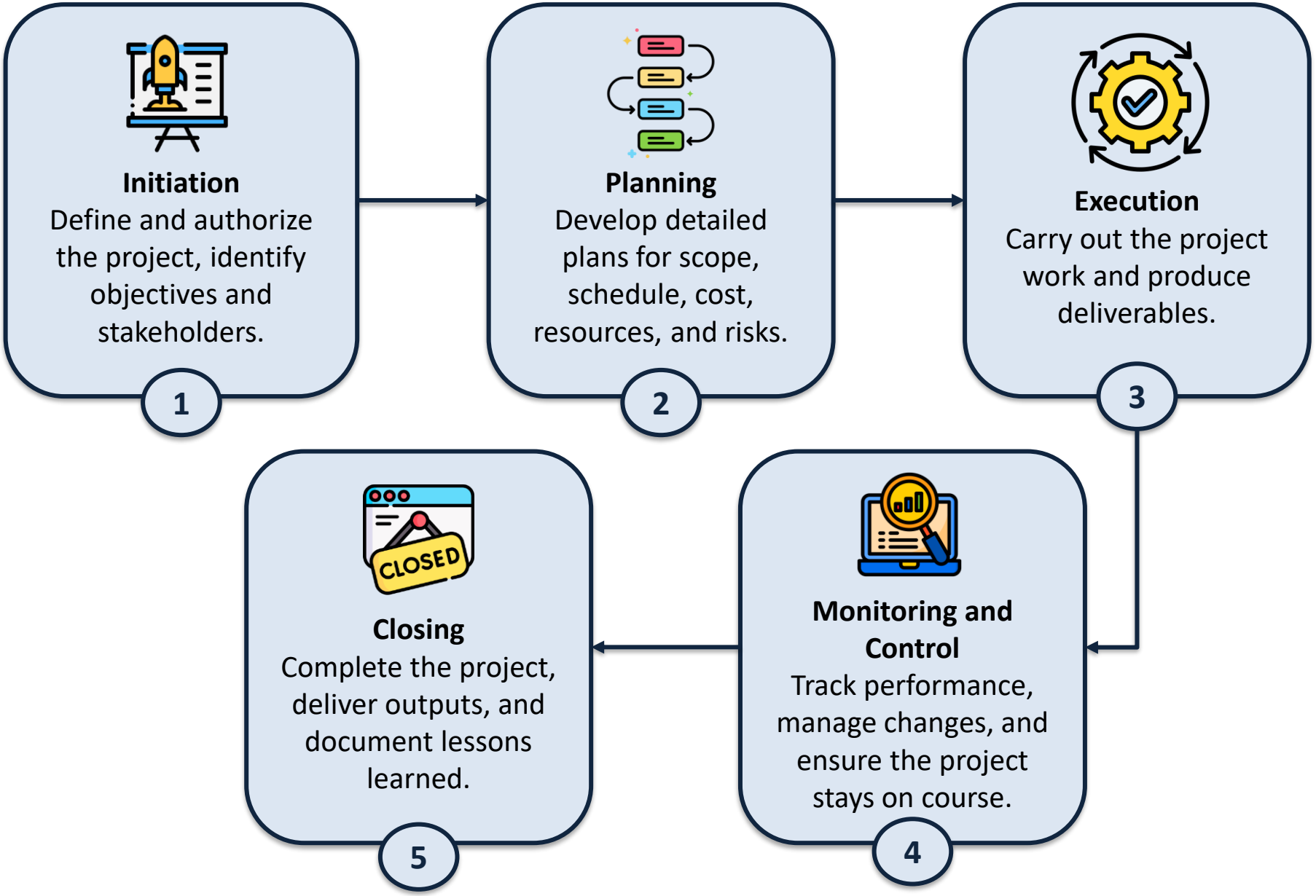
It gives a structured path for planning, executing, and delivering outcomes.

Key Aspects of Project Lifecycle

- Defined phases with specific objectives and deliverables.
- Provides structure and control for managing project work.
- Includes phase gates (review and approval points).
- Supports progressive development of project details.
- Can vary depending on project type (predictive, iterative, agile).



Typical Project Lifecycle Phases



Project Lifecycle Characteristics

- **Sequential or Overlapping Phases**

  Phases run in order or overlap, depending on the approach.
- **Defined Deliverables for Each Phase**

  Each phase produces its own specific outputs.
- **Use of Phase Gates (Review Points)**

  End-of-phase reviews decide whether to proceed, revise, or stop.
- **Uncertainty is Highest at the Beginning**

  Highest early on, when information is limited.
- **Risk and Stakeholder Influence is Higher Early On**

  Stakeholders can influence the project most in early phases.
- **Cost of Changes Increases Over Time**

  Changes get costlier as the project progresses.

IT Project Lifecycle Example



Requirements Gathering
Identify and document user needs, system requirements, and project objectives.

1



System Design
Define the system architecture, technologies, and overall solution design.

2



Development
Build and code the system components based on the design specifications.

3



Testing
Verify that the system works correctly by identifying and fixing defects.

4



Deployment
Release the system to users or the production environment.

5



Maintenance
Provide ongoing support, updates, and improvements after deployment.

6

In-Class Activity 2 – Multiple Choice Question

What is the correct order of typical project lifecycle phases?

A. Execution → Planning → Initiation → Closing

B. Initiation → Planning → Execution → Monitoring & Controlling → Closing

C. Planning → Initiation → Execution → Closing

D. Execution → Monitoring → Planning → Closing





Lifecycle vs Process Groups



- **Project Lifecycle**

  Refers to the **high-level phases** a project goes through from start to finish — it's the **overall structure** of the project.

- **Process Groups**

  Refer to the **set of activities or processes** performed to manage the project.

- **Key Differences**
  - Lifecycle focuses on **“when” work happens** (phases over time)
  - Process groups focus on **“how” work is performed** (management activities)
  - Lifecycle is **phase-based**, while process groups are **activity-based**
  - Process groups can **occur within each lifecycle phase** and may repeat
- The lifecycle provides the **structure**, while process groups provide the **actions needed to manage the project effectively**.

Introduction to Process Groups

- Process groups are **logical groupings of project management activities** used to guide a project from start to finish.
- There are **five main process groups**:

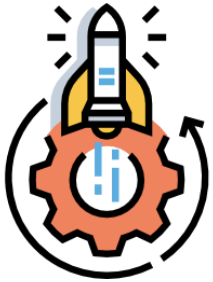


- Not the same as phases — they describe what actions happen, and when.
- Iterative and overlapping — they can repeat throughout the lifecycle.
- They ensure projects are properly started, planned, executed, controlled, and completed.



Process Group Interaction

- The **Process Group Interaction** is how the five process groups work **together and overlap**, not as a one-time sequence.
- Process groups are **interconnected and interdependent** — one group's output often feeds into another.
- They are **not strictly linear**; instead, they **overlap and interact continuously** during the project lifecycle.
- **Planning is iterative**, often revisited as new information or changes arise.
- **Monitoring & Controlling occurs throughout the project**, ensuring performance is tracked and adjustments are made when needed.
- **Execution and Planning often run in parallel**, especially in dynamic environments like IT projects.



Initiating Process Group

- Define and start the project
- Establish purpose & objectives
- Identify key stakeholders
- Define scope & requirements
- Secure authorization to begin
- Project charter & stakeholder register produced
- Sets project direction & foundation

Initiating Process Group Outputs



Project Charter

Authorizes the project, scope, objectives, and PM authority.



Stakeholder Register

Identifies stakeholders and their influence.

Together: **approval to start** + **who's involved & project goal**



Planning Process Group

- Defines scope, objectives, and course of action.
- Creates a roadmap for execution & control.
- Plans scope, schedule, cost, resources, quality, risk.
- Identifies issues early; prepares responses.
- Coordinates team & stakeholders.
- **Provides a structured, aligned plan.**

Planning Process Group Outputs



Project Management Plan: guides execution & control.



Scope Statement: project scope & deliverables.



Project Schedule: activities & deadlines.



Project Budget: estimated & approved cost.



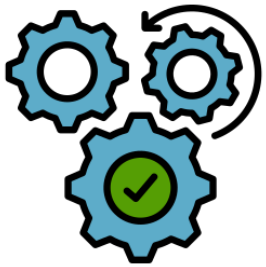
Risk Register: risks, impact, response.



Resource Management Plan: required resources.



Communication Plan: how info is shared.



Executing Process Group

- **Completes the work in the project plan.**
- **Produces project deliverables.**
- **Coordinates people, resources, tasks** toward objectives.
- **Manages team performance & communication.**
- **Maintains quality standards.**
- **Actual project work** is carried out to **create deliverables**.

Executing Process Group Outputs



Project Deliverables: outputs produced.



Work Performance Data: progress, time, cost data.



Issue Log: problems & resolutions.



Change Requests: requested modifications.



Project Documents Updates: updated project records.



Monitoring & Control Process Group

- **Tracks, reviews, and regulates performance.**
- Keeps the project **aligned with the plan.**
- **Identifies variances** from plan.
- **Implements corrective actions** as needed.
- **Manages changes, risks, issues**.
- Keeps it **on track to desired outcomes.**

Monitoring and Control Process Group Tools



Work Performance Reports: status & progress summary.



Change Requests: requests to modify scope.



Project Management Plan Updates: plan revisions.



Project Documents Updates: record updates.

Forecasts: predicted future performance.





Closing Process Group

- **Formally closes the project or phase.**
- **Confirms objectives are achieved.**
- **Deliverables accepted by stakeholders.**
- **Hands over outputs** to client/operations.
- Releases **resources**, no pending work.
- Project is **finalized and documented.**



Closing Process Activities

1. **Finalize Deliverables**

   Outputs completed & meet requirements.
2. **Obtain Stakeholder Acceptance**

   Formal approval from stakeholders.
3. **Transfer Deliverables**

   Hand over to **customers/operations**.
4. **Document Lessons Learned**

   Record **successes & improvements** for future projects.
5. **Close Contracts**

   Settle **vendor/supplier agreements**.
6. **Release Resources**

   Release **team & project resources**.
7. **Archive Project Documents**

   Store records for **future reference**.



Why Still Study Process Groups?

Even with a **principle- and performance-based approach**, process groups still matter:

- Clear structure for managing projects
- Widely used in industry
- Shows the overall project flow
- Key for certifications (e.g., PMP)
- Complements principles & domains
- Supports planning & control
- Applies to real-world IT projects

A practical guide that still supports effective execution today.

Agile vs Traditional Lifecycle

- **Traditional (Predictive) Lifecycle**

  A **sequential approach** where project phases are completed one after another. Requirements are defined early, and changes are more difficult and costly later in the project.

- **Agile (Adaptive) Lifecycle**

  An **iterative and incremental approach** where work is done in small cycles. Requirements can evolve, and feedback is incorporated continuously.

Agile vs Traditional : Key Differences		
	Agile	**Traditional**
Approach	Iterative	Sequential
Planning	Adapts continuously	Uses upfront planning
Delivery	Delivers in increments	Delivers at the end
Flexibility	Flexible to change	Rigid
Stakeholder Management	Continuous	Limited

Traditional lifecycle is best for **stable and well-defined projects**, while Agile is suitable for **dynamic and changing environments**, especially in IT projects.

What is a Performance Domain?

The 8 Performance Domains key areas of focus needed

to manage and deliver a project.

How they connect to Process Groups: process groups describe activities; domains focus on outcomes.

Together: a flexible framework focused on value, adaptability, and success.

Eight Performance Domains



Stakeholder Performance Domain

Identify, understand, engage stakeholders.



Team Performance Domain

Build & lead a high-performing team.



Development Approach and Life Cycle Performance Domain

development approach & lifecycle choice



Planning Performance Domain

Organize and coordinate project activities.

Eight Performance Domains



Project Work Performance Domain

Execute project work effectively.



Delivery Performance Domain

Deliver valuable outcomes & benefits.



Measurement Performance Domain

Track, assess, report performance.



Uncertainty Performance Domain

Manage risk and uncertainty.

Core Characteristics of Performance Domain

- **Focus on outcomes, not processes.**
- **Interrelated** areas working together.
- **Guide decision-making** & improvement.
- **Flexible and adaptable** across project types.
- **Support value & stakeholder satisfaction.**

Mapping Process Groups to Performance Domains

Shows how traditional process-based activities align with outcome-focused domains.

Process groups (activities) and performance domains (outcomes) complement each other.

Mapping Process Groups to Performance Domains

Process Group to Performance Domain Map		
Process Group	**Performance Domain**	**Description**
Initiating	Planning Performance Domain	Focuses on defining scope, schedule, resources, and overall project direction.
Planning	Project Work Performance Domain	Involves carrying out project tasks and producing deliverables.
Executing	Measurement Performance Domain	Tracks performance, evaluates progress, and supports decision-making.
Monitoring & Controlling	Stakeholder Performance Domain	Emphasizes identifying and engaging stakeholders early in the project.
Closing	Delivery Performance Domain	Focuses on delivering final outputs and ensuring value realization.

In-Class Activity 3 – Multiple Choice Question

Which process group focuses on tracking project performance and managing changes?

A. Executing

B. Planning

C. Monitoring & Controlling

D. Closing



Short Answer Question

What is one key difference between **process groups** and **performance domains**?





Key Challenges in IT Projects

- **Scope Creep**

  Uncontrolled requirement growth — delays, added cost.
- **Changing Requirements**

  Frequent updates complicate planning.
- **Resource Constraints**

  Limited people, time, or budget.
- **Technology Risks**

  New/complex tech — integration issues.
- **Communication Gaps**

  Poor communication — misunderstandings.
- **Quality Issues**

  Weak standards — defects & rework.
- **Uncertainty and Risk**

  High uncertainty makes risk management critical.

Best Practices in Project Management



Continuous Stakeholder Engagement

Keep stakeholders informed & involved.



Iterative Planning

Refine plans as new information emerges.



Effective Communication

Clear, timely, transparent — for everyone.

Best Practices in Project Management



Proactive Risk Management

Address risks early.



Focus on Value Delivery

Align work with real benefits.



Quality Management

Build quality in to cut rework.



Adaptability and Flexibility

Adjust plans as things change.

Case Study: Development of a National E-Learning Platform in Sri Lanka

Background:

The Ministry of Education Sri Lanka initiates a project to develop a **national e-learning platform** to support remote education for schools and universities, especially after disruptions like pandemics and natural disasters.

Project Objectives:

- Provide **online learning access** to students across the country.
- Enable **teachers to upload and manage content.**
- Support **live classes and assessments.**
- Ensure accessibility in **urban and rural areas.**



Project Lifecycle Application



Initiation: Define project goals, identify stakeholders (students, teachers, government), approve project.



Planning: Define system requirements, budget, timeline, risks, and resource needs.



Execution: Develop platform, create content, conduct training programs.



Monitoring & Controlling: Track progress, manage issues (e.g., connectivity problems), ensure quality.



Closing: Deploy system nationwide, obtain acceptance, document lessons learned.

Process Groups Application



Initiating: Project charter approved, stakeholders identified.



Planning: Scope, schedule, and risk plans developed.



Executing: Platform development and implementation.



Monitoring & Controlling: Performance tracking, issue resolution, change management.



Closing: Final delivery and stakeholder acceptance.

Performance Domains Application



Stakeholder: Engage teachers, students, IT teams, and policymakers.



Team: Coordinate developers, designers, and trainers.



Development Approach: Use a hybrid approach (Development - Agile, Rollout - Predictive).



Planning: Define timelines and resources.



Project Work: Execute development and deployment.



Delivery: Ensure platform meets educational needs.



Measurement: Track usage, performance, and adoption.



Uncertainty: Manage risks like internet access and system failures.



Key Challenges in This Project:

- Limited **internet access in rural areas**
- Frequent **requirement changes from stakeholders**
- **Resource constraints** (technical expertise, funding)
- **Technology adoption resistance** from teachers



Best Practices Applied:

- Continuous **stakeholder engagement** (teachers and students)
- **Iterative development** for platform features
- Strong **communication channels**
- Proactive **risk management** for connectivity issues
- Focus on **value delivery** (accessible education for all)

Summary

- **Project Management**: Applying skills and tools to **deliver value**
- **Lifecycle Phases**: Initiation → Planning → Execution → Monitoring & Controlling → Closing
- **Process Groups**: Initiating, Planning, Executing, Monitoring & Controlling, Closing (**activities across all phases**)
- **Key Difference**: Lifecycle = *structure* | Process Groups = *management activities*
- **12 Principles**: Guide **value, leadership, adaptability, stakeholder focus**
- **8 Performance Domains**: Focus on **key outcome areas** (stakeholders, team, delivery, uncertainty)
- **Interaction**: Process groups are **iterative and overlapping**
- **Agile vs Traditional**:

  Traditional = sequential | Agile = iterative & flexible
- **Challenges**: Scope creep, changing requirements, resource & technology constraints
- **Best Practices**: Stakeholder engagement, iterative planning, risk management, communication, value focus



References & Further Reading

The lifecycle and process-group concepts in this lecture are grounded in the following project management literature and industry certifications:

Recommended Books

- **A Guide to the Project Management Body of Knowledge (PMBOK® Guide)** – 7th Edition, Project Management Institute (PMI)
- **Schwalbe, K.** – *Information Technology Project Management*
- **Kerzner, H.** – *Project Management: A Systems Approach to Planning, Scheduling, and Controlling*

Certification Course Materials

- **PMI Project Management Professional (PMP)®** – process groups, knowledge areas
- **PMI Certified Associate in Project Management (CAPM)®** – entry-level PM fundamentals
- **AXELOS PRINCE2® Foundation and Practitioner** – process-based lifecycle method
- **Scrum Alliance Certified ScrumMaster (CSM)®** – agile delivery for IT projects

Students preparing for a career in IT project management are encouraged to explore these resources and certifications to build industry-recognised, practical expertise.